地质学专业培养方案

专业名称与代码: 070901

专业培养目标: 本专业培养品德端正、基础扎实、实践动手能力强的地质学及相关领域通用型高级人才。毕业生具有坚实的数理化基础和良好的人文素养，具有地球系统科学思维、较全面的地质学理论基础和较强的实践动手能力，熟悉常用的计算机软件，具备基本的英语听、说、读、写能力。具有严谨务实的科学精神，具备较强的自主学习能力及创新创业能力。本科毕业生能适应当前地球科学的发展和国家在资源、环境、灾害、国土规划以及国民经济其它相关领域对地质学人才的需要，部分毕业生继续攻读研究生后可成为科研机构、高等院校从事地球科学研究的菁英人才。

专业毕业要求:

1. 思想道德与人文素养：具有坚定的政治立场、良好的道德修养和正确的人生观与价值观；具备艰苦朴素、求真务实的专业精神和热爱自然、服务祖国的情怀。

2. 理科基础与思维能力：具有较好的数学、物理、化学、生物学和信息科学基础，具有一定的逻辑思维能力和分析解决问题的能力。

3. 专业知识体系：掌握地质学的基本理论和知识体系，初步具备地球系统科学思维，可认识常见岩石、构造和古生物等地质现象；能够基于学科专业相关背景知识，具有从事找矿勘探、灾害地质、工程地质、环境地质、地球化学、资源开发等方面的潜在工作能力，服务社会需求。

4. 实践与动手能力：掌握地质学的基本工作方法和常用技术手段，具有岩石矿物识别、古生物鉴定和构造分析等方面的基本技能，有较强的野外地质工作能力；了解学科前沿，具有较好的综合分析能力；初步具备从事地质学及相关方向工作和科学研究的能力。

5. 交流与团队合作能力：具有较好的理解和沟通能力，具备团队合作精神和一定的组织管理能力，能够就本学科的专业问题与业界同行及社会公众进行有效沟通和交流，具备基本的英语听、说、读、写能力和国际视野，能够在跨文化背景下进行沟通和交流。

6. 自主学习与创新意识：能够根据自身发展和社会需求积极开展自主学习；在了解学科前沿、国家目标及国际形势的基础上，积极开展创新性实践与探索。

毕业要求及实现途径：

序号	毕 业 要 求	实现途径（教学过程）
1	**思想道德与人文素养：** 具有坚定的政治立场、良好的道德修养和正确的人生观与价值观；具备艰苦朴素、求真务实的专业精神和热爱自然、服务祖国的情怀。	**①课堂教学：** 思想道德修养与法律基础、马克思主义基本原理、毛泽东思想和中国特色社会主义理论体系、习近平新时代中国特色社会主义思想概论、体育、军事理论、大学语文、传统文化及中西方文化选修课、哲学基础、人文艺术类选修课 **②课外学习：** 经典阅读与分享、人文讲座、网上主题团课、社会调查等
2	**理科基础与思维能力：** 具有较好的数学、物理、化学、生物学和信息科学基础，具有一定的逻辑思维能力和分析解决问题的能力。	**①课堂教学：** 高等数学、线性代数、概率论与数理统计、大学物理、物理实验、普通生物学、大学化学、化学实验、逻辑学等 **②课外学习：** 大学生数学建模、物理实验创新设计、化学实验创新设计等竞赛活动及相关的系列科普活动
3	**专业知识与技能：** 掌握地质学的基本理论和知识体系，初步具备地球系统科学思维，可认识常见岩石、构造和古生物等地质现象；能够基于学科专业相关背景知识，具有从事找矿勘探、灾害地质、工程地质、环境地质、地球化学、资源开发等方面的潜在工作能力，服务社会需求。	**①课堂教学：** 地球系统科学导论、地球化学、地球物理学概论等学科基础课；地球物质 I，II、地球表层系统科学、地球生物学、沉积地球系统科学等专业核心课；遥感地质学、工程地质学基础、海洋学基础、灾害地质学、环境科学导论、经济地理学、资源与环境经济学等跨学科/跨专业选修课；北戴河、周口店和秭归野外地质实习、海内外经典地质路线实习 **②课外学习：** “地球系统科学论坛”、“名家论坛”等系列学术报告与交流
4	**实践与动手能力：** 掌握地质学的基本工作方法和常用技术手段，具有岩石矿物识别、古生物鉴定和构造分析等方面的基本技能，有较强的野外地质工作能力；了解学科前沿，具有较好的综合分析能力；初步具备从事地质学及相关方向工作和科学研究的能力。	**①课堂教学：** 地球物质、地球生物学、构造地质学等专业课实习；北戴河、周口店和秭归三大野外地质实习、海内外经典地质路线实习、Python 语言程序课程设计、地学大数据分析与应用、人工智能等相关课程 **②课外学习：** 参加大学生科技论文报告会、大学生创新创业训练计划项目;参观国家重点实验室、博物馆等；参与课余科研立项
5	**交流与团队合作能力：** 具有较好的理解和沟通能力，具备团队合作精神和一定的组织管理能力，能够就本学科的专业问题与业界同行及社会公众进行有效沟通和交流，具备基本的英语听、说、读、写能力和国际视	**①课堂教学：** 野外地质分组实习与团队合作；大学英语、部分学科基础课和专业核心课双语教学等 **②课外学习：** 外文文献阅读、大学生英语竞赛、国外名家学术讲座、全国性学术会议、与本校留学生交流活动、世界名校访学交流等

序号	毕　业　要　求	实现途径（教学过程）
	野，能够在跨文化背景下进行沟通和交流。	
6	**自主学习与创新意识：** 能够根据自身发展和社会需求积极开展自主学习；在了解学科前沿、国家目标及国际形势的基础上，积极开展创新性实践与探索。	**①课堂教学：** 以问题为导向的专业课程教学、以学生为主体的研讨式课堂教学 **②课外学习：** 团队化科研训练、实践能力训练、社会调查、“赛恩师”科技领航系列活动、“互联网+”、“挑战杯”等重大科创赛事

主干学科: 地质学

专业核心课程: 地球系统科学导论、地球物质 I，II、地球生物学、沉积地球系统科学、构造地质学、地貌学与第四纪地质学、地球物理学概论、地球化学、矿床学

主要专业实验: 晶体模型定向与分析、矿物标本鉴定、岩石手标本与薄片鉴定、古生物鉴定、沉积地质实习、构造分析

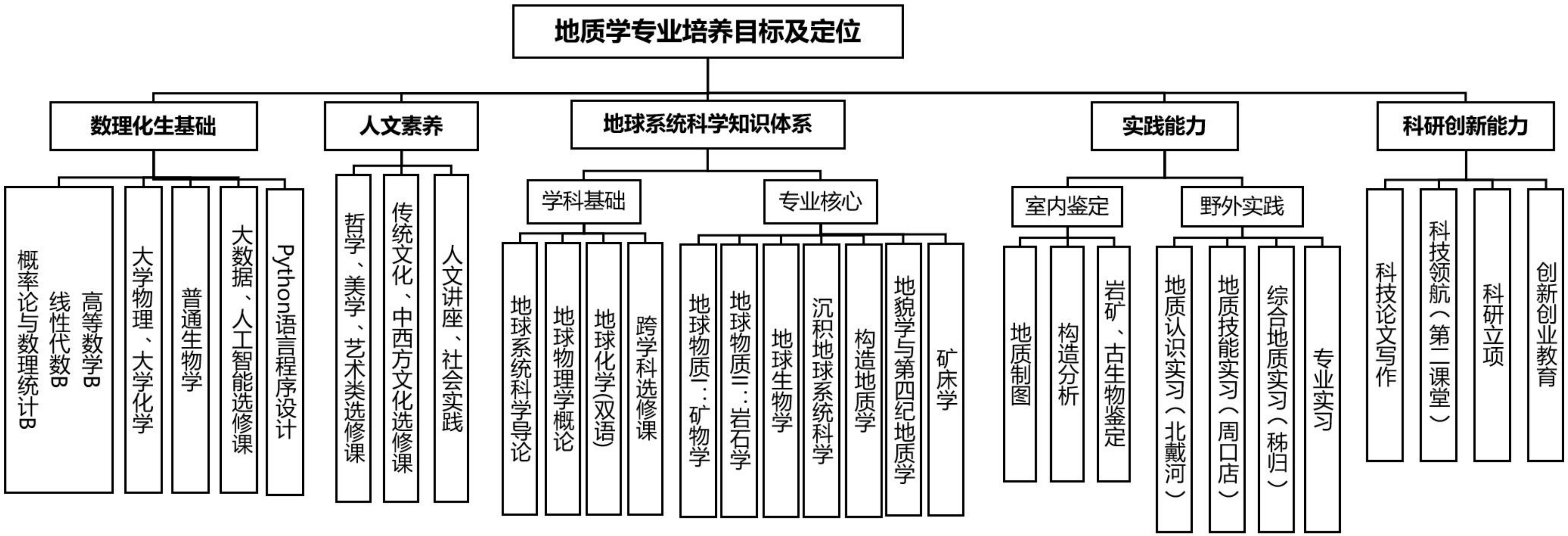
主要实践性教学环节: 地质认识实习(北戴河)、地质技能实习（周口店）、综合地质实习（秭归），科研立项，Python 语言程序设计、毕业生产实习、毕业论文（设计）

毕业学分要求: 165 学分

学制与学位: 四年，理学学士

本专业学生可以辅修的其他专业: 地理科学、地质工程、海洋科学、环境工程、大气科学

相近专业: 地球化学、地球物理学


地质学专业培养目标及定位
数理化生基础
人文素养
地球系统科学知识体系
实践能力
科研创新能力
高等数学B
线性代数B
概率论与数理统计B
大学物理、大学化学
普通生物学
大数据、人工智能选修课
Python语言程序设计
哲学、美学、艺术类选修课
传统文化、中西方文化选修课
人文讲座、社会实践
学科基础
专业核心
地球系统科学导论
地球物理学概论
地球化学(双语)
跨学科选修课
地球物质一：矿物学
地球物质二：岩石学
地球生物学
沉积地球系统科学
构造地质学
地貌学与第四纪地质学
矿床学
室内鉴定
野外实践
地质制图
构造分析
岩矿、古生物鉴定
地质认识实习（北戴河）
地质技能实习（周口店）
综合地质实习（秭归）
专业实习
科技论文写作
科技领航（第二课堂）
科研立项
创新创业教育

Program for Geology Major

Specialty and Code: Geology 070901

Education Objectives: This major aims to cultivate senior talents in geology and related fields with good moral character, solid foundation and strong practical ability. Graduates should have a solid foundation in mathematics, physics and chemistry, good humanistic quality, scientific thinking of the Earth system, comprehensive theoretical basis of geology and strong practical ability, familiar with commonly used computer software, and basic English listening, speaking, reading and writing ability. Graduates should have a rigorous and pragmatic scientific spirit, with strong self-learning ability and innovation and entrepreneurship ability. Graduates can adapt to the current development of geoscience and the needs of the country for geological talents in the fields of resources, environment, geohazards, territorial planning and other related fields of national economy. Some graduates can become elite talents engaged in geoscience research in scientific research institutions and universities after continuing to study for graduate students.

Graduation Requirements:

1. Moral & humanistic quality: Have a firm political position, good moral cultivation and correct outlook on life and values; have the professional spirit of hard work and plain living, seeking truth and pragmatism, and dedication to serving the motherland.

2. Scientific literacy and critical thinking: Have a good foundation in mathematics, physics, chemistry, biology and information science and the ability to analyze and solve problems, with strong logical thinking ability.

3. Professional foundation and skills: Grasp the basic theory and knowledge system of geology, have a basic vision of Earth system science, can understand common geological phenomena such as rocks, structures and paleontology; have the potential ability to engage in prospecting and exploration, geohazards, engineering geology, environmental geology, geochemistry, resource development and other aspects to serve the needs of society.

4. Practical ability: Master the basic methods and common technical means of geology, have the basic skills of rock and mineral identification, paleontology identification and structural analysis, have strong field geological work ability; understand the frontier of disciplines, have good comprehensive analysis ability; have the preliminary ability to engage in geology and related work and scientific research.

5. International Vision and Intercultural Communication: Have good communication skills, possess team spirit and organizational skills, be able to effectively communicate and exchange with industry peers and the public on professional issues of the discipline, have basic English listening, speaking, reading and writing skills and international vision, and be able to communicate and exchange in a cross-cultural context.

6. Independent Learning and Creativity: Be able to actively engage in independent study in accordance with their own development and the needs of society; can actively engage in innovative practice and exploration on the basis of an understanding of the frontiers of geosciences, national goals and the international situation.

Graduation Requirements and Pathways:

No.	Graduation Requirements	Pathways（Teaching Process）
1	**Moral & humanistic quality:** Have a firm political stance and a strong sense of social responsibility; have a correct outlook on life and values; possess a professional spirit of hard work, simplicity, and pragmatism.	**①Classroom Teaching:** Ideological and Moral Cultivation and legal Foundation, Situation and Policy, Introduction to Mao Zedong Thought and the Theoretical System of Socialism with Chinese Characteristics, Labor Practice, College Chinese, Traditional Culture and Chinese and Western Culture Elective Courses, Philosophy Foundation, Aesthetics and Art Elective Courses **②Extracurricular Learning:** Classic reading and sharing, humanities lectures, social surveys, etc.
2	**Scientific literacy and critical thinking:** Have a solid foundation in mathematics, physics, chemistry, biology and information science and the ability to analyze and solve problems, with strong logical thinking ability.	**①Classroom Teaching:** Advanced Mathematics and , Linear Algebra and , Probability Theory and Mathematical Statistics and Geosciences Application, College Physics and Geosciences Application, Physics Experiment, General Biology, College Chemistry and , Chemistry Experiment, Logic, etc. **②Extracurricular Learning:** Competition activities such as Mathematical Modeling, Innovative Design of Physical Experiments, Innovative Design of Chemical Experiments, and a series of popular science activities
3	**Professional foundation**: Establish a basic knowledge system of Earth materials, geochemistry, geophysics, Earth observation, Earth surface system, Earth environment and life evolution, and have the vision of Earth system science; have the basic knowledge and skills to engage in related work in various fields of geosciences.	**①Classroom Teaching:** Introduction to Earth System Science, Geochemistry, Introduction to Geophysics and other basic courses; Geomaterials I, II, Earth Surface System Science, Geobiology, Sedimentary Earth System Science and other professional core courses; Interdisciplinary elective courses such as Remote Sensing Geology, Fundamentals of Engineering Geology, Fundamentals of Oceanography, Geohazard Science, Introduction to Environmental Science, Economic Geography,

No.	Graduation Requirements	Pathways（Teaching Process）
		Resources and Environmental Economics, etc.; field geological practices in Beidaihe, Zhoukoudian and Zigui, and practice of classic geological routes ②**Extracurricular Learning:** Series of academic reports and exchanges such as "Earth System Science Forum" and "Famous Experts Forum"
4	**Research and practical ability**: Master the basic methods and common technical means of geology, have the basic skills of rock and mineral identification, paleontology identification and structural analysis, have strong field geological work ability; understand the frontier of disciplines, have good comprehensive analysis ability; have the preliminary ability to engage in geology and related work and scientific research.	①**Classroom Teaching:** Beidaihe and Zhoukoudian Field Geological Practice, Classic Geological Route Practice at Home and Abroad, Scientific Paper Writing, Python Language Program Course Design, Artificial Intelligence and other related courses ②**Extracurricular Learning:** Scientific research projects: joining the tutor's scientific research team and participated in scientific research projects, organizing teams freely to start innovative research projects, and participate in 1-2 academic conferences; college students 'scientific and technological thesis reports, and college students' innovation and entrepreneurship training programs.
5	**International Vision and Intercultural Communication**: Have good communication skills, team spirit and organizational skills, be able to effectively communicate and exchange with industry peers and the public on professional issues of the discipline, have basic English listening, speaking, reading and writing skills and international vision, and be able to communicate and exchange in a cross-cultural context.	①**Classroom Teaching:** College English; Bilingual teaching of selected core courses; field practice in Beidaihe, Zhoukoudian and Zigui, domestic and overseas geological trips ②**Extracurricular Learning:** Reading of literature in a foreign language, academic lectures by foreign experts, exchange activities with international students, popular science activities, overseas study tours, etc.
6	**Independent Learning and Creativity**: Be able to actively engage in independent study in accordance with their own development and the needs of society; to actively engage in innovative practice and exploration on the basis of an understanding of the frontiers of geosciences, national goals and the international situation.	①**Classroom Teaching:** Problem-oriented classroom teaching, student-centered seminar discussion ②**Extracurricular Learning:** Scientific research training, practical ability training, social investigation, discipline competition, academic report and discussion, etc.

Major Disciplines: Geology

Main Courses: Introduction to Earth System Science, Earth Materials (I, II), Geobiology, Sedimentary Earth System Science, Structural Geology, Geomorphology and Quaternary Geology, Geochemistry, Geophysics, Mineral Deposits

Lab Experiments: Identification of Crystal Structure Models, Identification of Minerals and Rocks, Fossil Identification, Tectonic Analysis.

Practical Work: Primary Field Training (Beidaihe in Hebei Province, North China), Geological Mapping Training (Zhoukoudian in Beijing), Geological Field Training (Zigui), Professional Fieldwork Training, Practice for Graduation, Thesis for Graduation.

Credits Required for Graduation: 165

Duration& Degree Granted: Four years, Bachelor of Science

Recommended Minors: Geography Science, Geological Engineering, Atmospheric science, Marine Science, Environmental Engineering

Related Specialties: Geochemistry, Geophysics

地质学专业课程教学计划表

Course Descriptions of Geology Major

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
						课内学时		课外学时											
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp		一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
通识教育课 Liberal Education Courses	必修 Compulsory	1200 7800	马克思主义基本原理 Principles of Marxism	3	48	48						3							
		1200 8100	毛泽东思想和中国特色社会主义理论体系概论 Introduction to Mao Tse-tung thought & the Theoretical System of Socialism with Chinese Characteristics	2	32	32									2				
		1200 8000	习近平新时代中国特色社会主义思想概论 Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3	48	48									3				
		1171 1800	中国近现代史纲要 The Essentials of Modern Chinese History	2	32	32							2						
		1200 7900	思想道德与法治 Ideological Morality & Rule of Law	3	48	32	16					3							
		1200 5300	形势与政策 Situation & Policy	2	32	32						每学期平均分配							
		1130 76*0	体育 Physical Education	4	144	144						1	1	1	1				
		1092 34*0	大学英语 College English	9	144	144				48		3	3	3					
		1430 0300	军事理论 Military Theory	2	36	36						2							
		1200 8200	劳动教育 Labor Education	1	16	16						1							
		2162 4800	大学语文 B College Chinese B	**2**	32	32							2						
		2012 3000	逻辑学导论 Logic	**2**	32	32						2							

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours 课内学时 讲课 Lec.	课内学时 课内实验 Lab	课外学时 实验/科研实践 Lab/Res.	课外学时 研讨 Dis	课外学时 素质拓展 Exp	先修课程 Prerequisite courses	学期学分分配 Semester Credits 一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
	2180 1000	哲学基础 Foundation of Philosophy	2	32	32						2							
选修 Elective		总计 12 学分。包括经典阅读与分享、人文讲座必修课程（各 1 学分），美育、心理健康教育课程（各不少于 2 学分），跨学科选修课不低于 4 学分。	12	192	192			32			4					4	4	
		人文讲座（1-7 学期） Humanities Seminars	1					16									1	
		经典阅读与分享（1-7 学期） Literature Reading & Sharing	1					16									1	
	小计 Sum		**49**	**868**	**852**	**16**		**32**	**48**		**21**	**8**	**4**	**6**		**4**	**5**	**1**
理科基础 Fundamentals of Science	2194 7400	Python 语言程序设计 Python Language Programming	2.5	40	**40**		16						2.5					
	2121 27*2	高等数学 B Advanced Mathematics B	10	160	160						4	6						
	2121 2802	线性代数 B Linear Algebra B	2.5	40	40									2.5				
	2121 3502	概率论与数理统计 B Probability Theory & Mathematical Statistics B	2.5	40	40										2.5			
	2121 30*1	大学物理 A College Physics A	8	128	128							4	4					
	2121 69*1	物理实验 A Physical Experiments A	2	64	4	60						1	1					
	2032 69*1	大学化学 A College Chemistry A	5.5	88	88							2.5	3					
	2032 70*1	大学化学实验 A College Chemistry Experiment A	2	48		48						1	1					
	2012 4400	科技论文写作 Science & Technology Paper Writing	1	16	16										1			
	小计 Sum		**36**	**624**	**516**	**108**	**16**				**4**	**14.5**	**11.5**	**2.5**	**3.5**			

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours 课内学时 讲课 Lec.	课内实验 Lab	课外学时 实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp	先修课程 Prerequisite courses	学期学分分配 Semester Credits 一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
学科基础课 Disciplinary Fundamental Courses	学科内必修	2012 3500	地球系统科学导论 Introduction to Earth System Science	3	48	36	12			16		3							
		2012 0200	地球化学 Geochemistry	3	48	48			16		岩石学导论、构造地质学、沉积地球系统科学				3				
		2062 8000	地球物理学概论 Introduction to Geophysics	3	48	44	4				高数、大学物理、线性代数					3			
		2040 6900	普通生物学 General Biology	2	32	32						2							
		累计		**11**	**176**	**164**	**12**		**16**	**16**		**5**			**3**	**3**			
	跨学科选修	**在学术导师指导下选择不少于10个学分的跨学科选修课**																	
		2230 3700	人工智能 Artificial Intelligence	2	32	28	4										2		
		2113 0402	测量学 B Surveying B	1.5	24	24		12				1.5							
		2114 8500	地图学 Cartography	2	32	32		8									2		
		2011 1601	自然地理学 A Physical Geography A	3	48	48										3			
		2011 7700	遥感地质学 Remote Sense Geology	3	48	32	16										3		
		2020 2800	海洋学基础 Basic Oceanography	3	48	48	0												3
		2012 1600	大气科学概论 Introduction to Atmospheric Science	1.5	24	24	0											1.5	
		2110 0700	GIS 原理与应用 Principle & Application of GIS	2.5	40	24	16											2.5	
		2050 8400	工程地质学基础 B Engineering Geology B	2.5	40	32	8												2.5
		2022 2303	石油及天然气地质学 C Petroleum Geology C	2.5	40	40		8									2.5		
		2023 0802	矿产勘查理论与方法 B Theories Method of Mineral Resource	2.5	40	40		8										2.5	
		2040 5303	环境评价 C Environmental Assessment C	2	32	32													**2**

地质学专业

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours 课内学时 讲课 Lec.	课内实验 Lab	课外学时 实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp	先修课程 Prerequisite courses	学期学分分配 Semester Credits 一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
	2170 4500	环境法规 Environmental Law	1.5	24	24													1.5
	2050 6000	地质灾害防治 Geological Hazards Control	2	32	32					工程地质基础						2		
	2193 4300	地学大数据分析与应用 Big Data Analysis & Application	2	32	32								2					
	2192 4300	云计算与大数据处理技术 Cloud Computing & Big Data Analysis	2	32	32					计算机网络B							2	
	2044 2100	大气数值模拟 Atmospheric Numerical Simulation	1	16	16					大气科学概论							2	
	2083 2900	资源与环境经济学 Resource & Environmental Economics	2	32	32											2		
	2054 9200	灾害地质学 Geohazard Science	2	32	32												2	
	2040 4400	环境规划与管理 Environment Planning and Management	2	32	32												2	
	2011 3200	区域分析与规划学 Regional Analysis and Planning	2	32	32												2	
	累计		**10**	**160**	**160**								**2**	**0**	**2**	**2**	**2**	**2**
	小计 Sum		**21**	**336**	**324**	**12**		**16**	**16**		**5**		**2**	**3**	**5**	**2**	**2**	**2**
专业核心课 Specialty Core Courses	2012 3710	地球物质 I：矿物学 Earth Materials I: Mineralogy	4	64	28	36	4			大学化学		4						
	2012 3720	地球物质 II：岩石学 Earth Materials II: Petrology	4	64	30	34	4	4		地球物质 I			4					
	2012 3800	地球生物学 Geobiology	3	48	32	16		8	8	地球系统科学导论			3					
	2012 3900	沉积地球系统科学 Sedimentary Earth System Science	4	64	48	16		8	8	地球生物学、地球物质 II、构造地质学				4				
	2010 4001	构造地质学 A Structural Geology & Tectonics A	4	64	40	24				地球系统科学导论、地球物质 II				4				
	2012 4500	地貌学与第四纪地质学 Geomorphology & Quaternary Geology	2.5	40	30	10			8						2.5			

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
					课内学时		课外学时											
					讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp		一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
	20222502	矿床学 B Mineral Deposits B	2	32	32		16			岩石学导论					2			
	小计 Sum		**23.5**	**376**	**240**	**136**	**16**	**20**	**32**		**0**	**4**	**7**	**8**	**4.5**	**0**	**0**	**0**
专业选修课 Specialty Elective Courses	在学术导师指导下选择不少于 9 个学分的专业选修课																	
	20116300	区域地质调查技术与方法 Methods & Techniques in Regional Geological Survey	2	32	16	16										2		
	20113800	全球变化 A Global Change A	3	48	40	8										3		
	20119200	行星地质学导论 Introduction to Planetary Geology	3	48	40	8									3			
	20124800	应用矿物学 Applied Mineralogy	3	48	32	16										3		
	20124600	岩石学前沿与实践 Frontier & Practice of Petrology	3	48	24	24	16								3			
	20124900	应用地球化学 Applied Geochemistry	2	32	32					地球化学						2		
	20118900	地震地质学 Earthquake Geology	3	48	48										3			
	20125000	地质学虚拟现实方法与应用 Virtual Reality Methods & Applications in Geology	2	32	16	16										2		
	20125100	遥感数字图像处理原理与应用 Principles & Applications of Remote Sensing Digital Image Processing	2	32												2		
	小计 Sum		9	144	136	8	16								3	3	3	
合计 Sub-total			138.5	2348	2068	280	56	68	88		30	26.5	24.5	19.5	16	9	10	3
实践环节 Practical Work	44300400	军事训练 Military Training	2	2 周							2							
	42008300	劳动实践 Labor Practice	1	4 周												1		
	41947500	Python 语言课程设计 Course Design for Python Language	1.5	1.5 周									1.5					
	40115200	地质认识实习（北戴河） Primary Field Training in Beidaihe	2	2 周								2						
	40124000	地质技能实习（周口店） Geological Field Training in Zhoukoudian	4	4 周										4				

地质学专业

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
					课内学时		课外学时											
					讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp		一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
	40125200	综合地质实习（秭归） Advanced Geological Field Training in Zigui	2	2 周												2		
	40118500	毕业生产实习 Practice for Graduation	4	4 周													4	
	40125500	毕业论文（设计） Thesis for Graduation	4	4 周														4
	小计 Sum		**20.5**	**23.5 周**							**2**	**2**	**1.5**	**4**	0	**3**	**4**	**4**
创新创业自主学习 Freedom study	ZZ35000S	社会调查 Social Investigation	2	2 周											2			
		科研立项 Scientific Research	2	4 周												2		
		其他(**含创新创业课程**、学科竞赛、发明创造、科研报告) Others (Start-up, Contest, Invention, Innovation & Research Presentation)	2	2 周														2
	小计 Sum		**6**	**8 周**							**0**	**0**	**0**	**0**	**2**	**2**	**0**	**2**
总计 Total			**165**	**2348+31.5 周**	**2068**	**280**	**56**	**68**	**88**		**32**	**28.5**	**26**	**23.5**	**18**	**14**	**14**	**9**

注: 全英课程须在课程名称后打*标出，通识教育选修课学分未列入具体学期，学院须根据学校创新创业自主学习学分认定一览表制订实施细则。

地质学专业课程分类统计

Course Classifications and Statistics of Geology

课程类别 统计	通识教育课程 Liberal Education Courses		理科基础课 Platform Disciplinary Fundamental Courses	学科基础课 Disciplinary Fundamental Courses		专业核心(核心)课 Main Specialty Courses	专业选修课 Specialty Elective Courses	实践环节 Practical Work	创新创业自主学习 Freedom Study	学时总计 Total Hours	学分总计 Total Credits
	必修	选修	必修	必修	选修						
课内总学时/学分	676/37	192/12	624/36	176/11	160/10	376/23.5	144/9	23.5 周/20.5	8 周/6	2348+31.5.周	165
课内总学时/学分	868/49			236/21							
学分所占比例	29.7%		21.8%	12.8%		14.2%	5.5%	12.4%	3.6%		100%

地质学专业辅修课程教学计划表

Course Descriptions of Geology（Minor）

课程类别 Category	课程编号 Code	课程名称 Course Name	学分 Crs	课内总学时 Hrs	学时分类 Class Hours 课内学时 讲课 Lec.	课内实验 Lab	课外学时 实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp	先修课程 Prerequisite courses	学期学分分配 Semester Credits 一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
学科基础课 Disciplinary Fundamental Courses	20119600	地球系统科学导论 Introduction to Earth System Science	2.5	40	40		8				2.5							
	20222502	矿床学 B Mineral Deposits B	2	32	32		16										2	
	20603800	固体地球物理学概论 Geophysics	2.5	40	40												2.5	
	小计 Sum		**7**	**112**	**112**	**0**	**24**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**2.5**	**0**	**4.5**	**0**
专业核心课 Main Specialty Courses	20113100	矿物岩石学 Mineralogy & Petrology	**3**	48	12	36									3			
	20118300	地球生物学 Geobiology	**3**	48	32	16										3		
	20104001	构造地质学 B Structural Geology B	**3**	48	30	18										3		
	20120200	地球化学 Geochemistry	3	48	48												**3**	
	20115100	地貌学与第四纪地质学 Geomorphology & Quaternary Geology	3	48	48													3
	小计 Sum		**15**	**240**	**170**	**70**					**0**	**0**	**0**	**0**	**3**	**6**	**3**	**3**
实践环节 Practical Work	40115200	地质认识实习(北戴河)	2	2 周												**2**		
	小计 Sum		**2**	2 周							**0**	**0**	**0**	**0**	**0**	**2**	**0**	**0**
总计 Total			**24**	**352**	**282**	**70**	**24**			**0**	**0**	**0**	**0**	**0**	**5.5**	**8**	**7.5**	**3**

地质学辅修专业课程分类统计

Course Classifications and Statistics of Geology（Minor）

课程类别 统计	学科基础课 Disciplinary Fundamental Courses	专业主干课 Main Specialty Courses	实践环节 Practical Work	学时总计 Total Hours	学分总计 Total Credits
课内总学时/学分	112/7	240/15	2 周/2	352+2 周	24
学分所占比例	29.2%	62.5%	8.3%		100%